Use of force-controlled grippers

1 Slider control

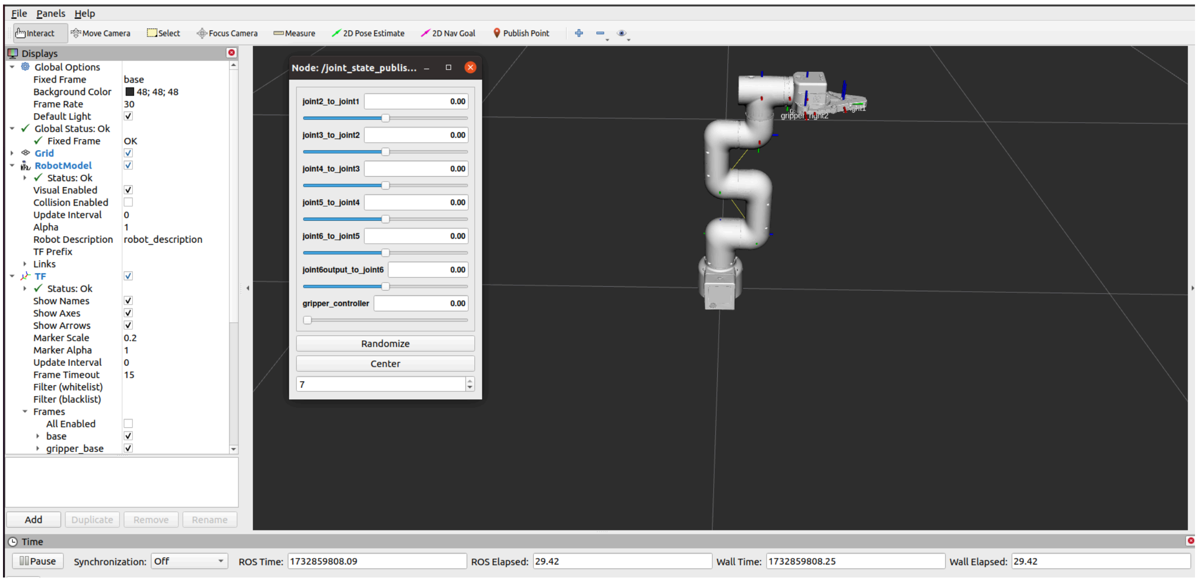
Open a command line and run:

> **Note: If the end effector uses myGripper F100 force-controlled gripper, the version of the pymycobot driver library must be greater than 3.6.4**

```
# If the end is equipped with a myGripper F100 force-controlled gripper, run (the default serial port name for the 2022 mycobot 320-M5 version is "/dev/ttyUSB0" and the baud rate is 115200. Some models have the serial port name "dev/ttyACM0". If the default serial port name is incorrect, you can change the serial port name to "/dev/ttyACM0"):
roslaunch new_mycobot_320 mycobot_320_force_gripper_slider.launch port:=/dev/ttyACM0 baud:=115200
```

Open **rviz and a slider component** , and you will see the following interface:

If the myGripper F100 force-controlled gripper is installed at the end, the following interface will be displayed:



Then you can **control the model** in rviz and move it by dragging the slider . If you want the real mold robot to move with the model, you need to open another command line and run:

> **Note: If the end effector uses myGripper F100 force-controlled gripper, the version of the pymycobot driver library must be greater than 3.6.4**

```
# If the end is equipped with a myGripper F100 force-controlled gripper, run (the default serial port name for the 2022 mycobot 320-M5 version is "/dev/ttyUSB0" and the baud rate is 115200. Some models have the serial port name "dev/ttyACM0". If the default serial port name is incorrect, you can change the serial port name to "/dev/ttyACM0"):
rosrun new_mycobot_320 mycobot_320_force_gripper_slider.py _port:=/dev/ttyACM0 _baud:=115200
```

Please note: Since the robot will move to the current position of the model when the command is entered, please make sure that the model in rviz does not appear to be intersecting before you use the command. Do not drag the slider quickly after connecting the robot arm to prevent damage to the robot arm.

2 Use of moveit

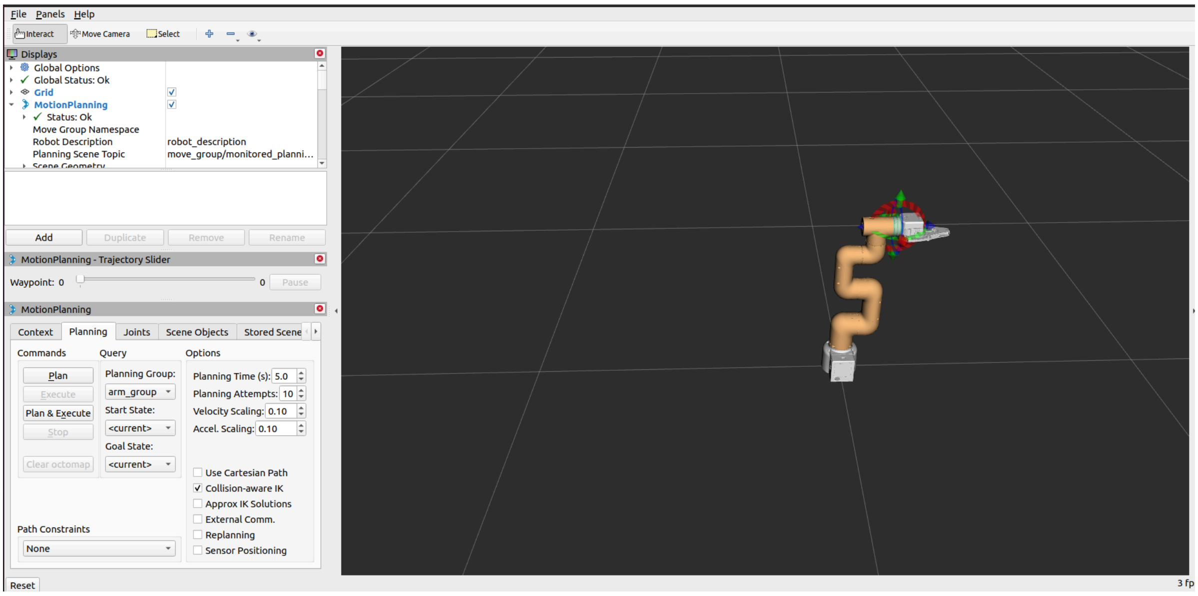
`mycobot_ros` integrates MoveIt parts.

> > **Note: If the end effector uses myGripper F100 force-controlled gripper, the version of the pymycobot driver library must be greater than 3.6.4**

The operation results are as follows:

```
# If the myGripper F100 force-controlled gripper is installed on the end, run:
roslaunch new_mycobot_320_force_gripper_moveit
mycobot320_force_gripper_moveit.launch
```

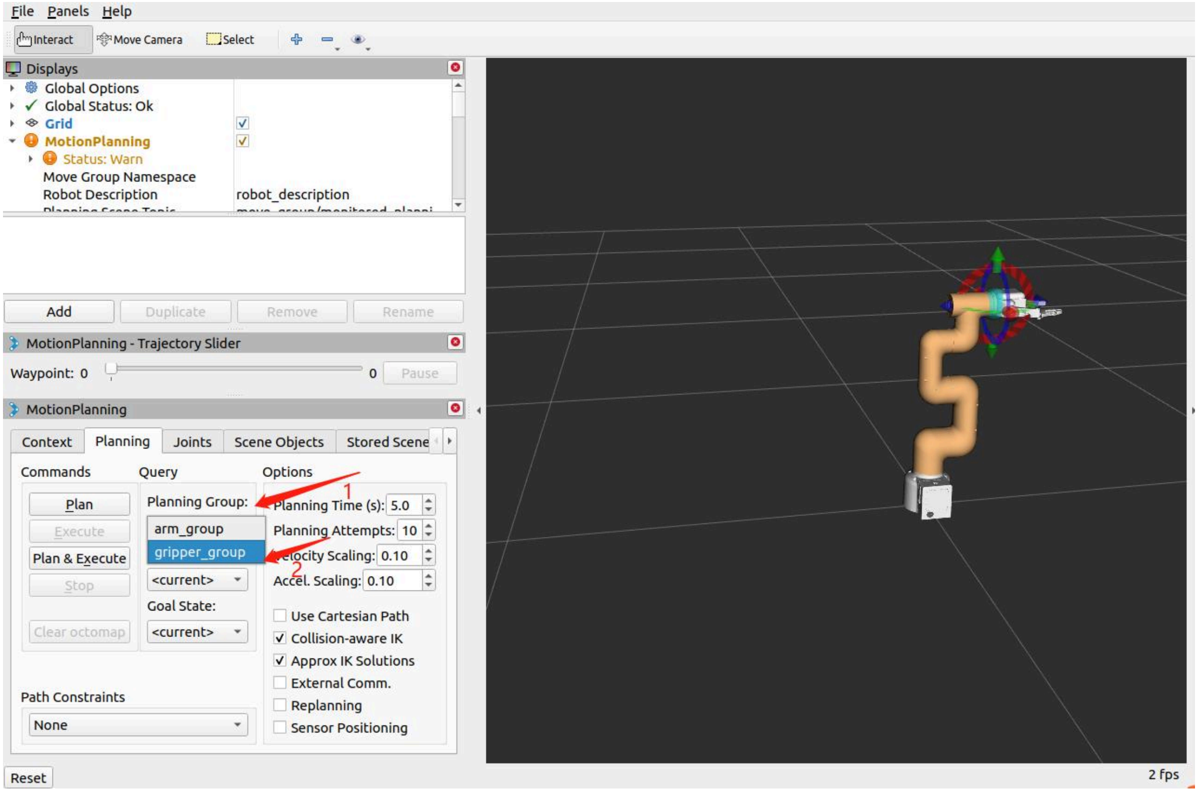
If the myGripper F100 force-controlled gripper is installed at the end, the operation effect is as follows:



If you want the real robot to execute the plan synchronously, you need to open another command line and run:

```
# If the end is equipped with a myGripper F100 force-controlled gripper, run (the default serial port name for the 2022 mycobot 320-M5 version is "/dev/ttyUSB0" and the baud rate is 115200. Some models have the serial port name "dev/ttyACM0". If the default serial port name is incorrect, you can change the serial port name to "/dev/ttyACM0"):
rosrun new_mycobot_320_force_gripper_moveit sync_plan.py _port:=/dev/ttyACM0
_baud:=115200
```

Note: If the end effector is equipped with an adaptive gripper or a myGripper F100 force-controlled gripper and the gripper needs to be planned, the planning group needs to be switched to the planning group of the gripper.


File Panels Help
Interact
Move Camera
Select
Displays
Global Options
Global Status: Ok
Grid
MotionPlanning
Status: Warn
Move Group Namespace
Robot Description
robot_description
Add
Duplicate
Remove
Rename
MotionPlanning - Trajectory Slider
Waypoint: 0
0
Pause
MotionPlanning
Context
Planning
Joints
Scene Objects
Stored Scene
Commands
Query
Options
Plan
Execute
Plan & Execute
Stop
Clear octomap
Planning Group:
arm_group
gripper_group
<current>
Goal State:
<current>
1
2
Planning Time (s): 5.0
Planning Attempts: 10
Velocity Scaling: 0.10
Accel. Scaling: 0.10
Use Cartesian Path
Collision-aware IK
Approx IK Solutions
External Comm.
Replanning
Sensor Positioning
Path Constraints
None
Reset
2 fps